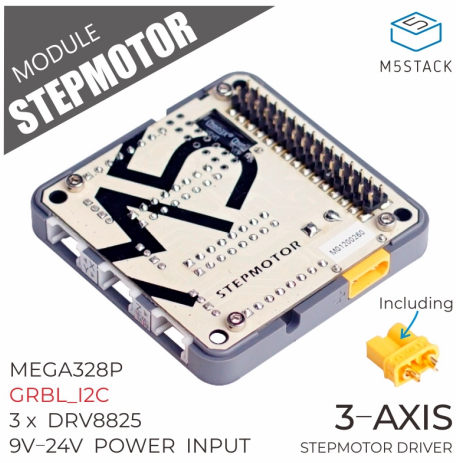
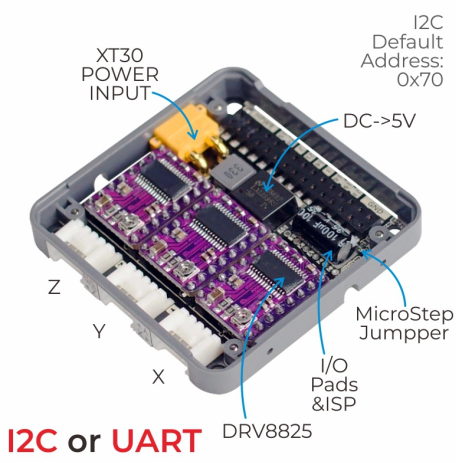
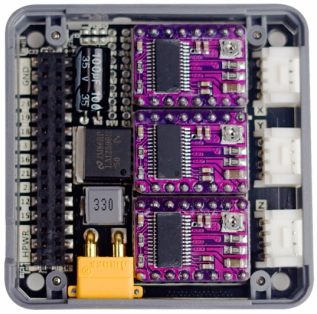
Module Stepmotor

SKU:M012

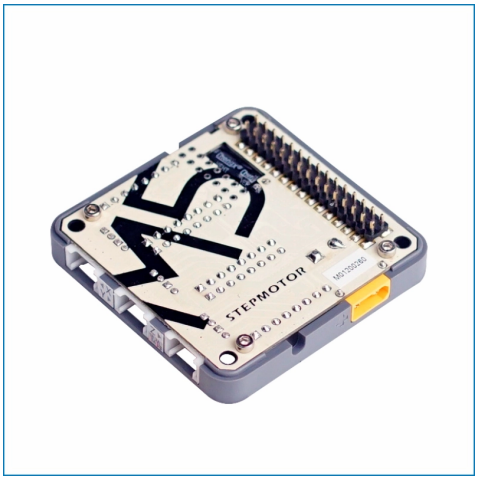






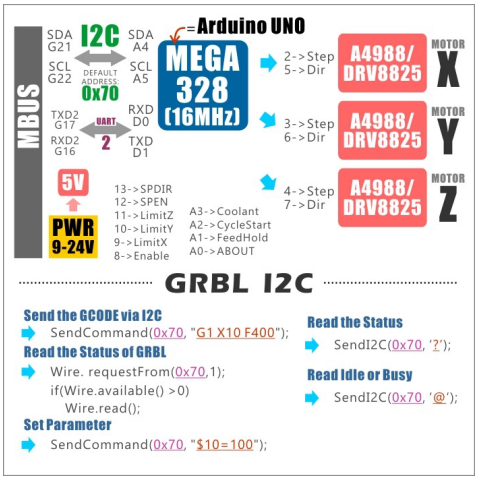












Description

Module Stepmotor is a stepper motor driver module in the M5Stack stackable module series. This module can drive 3 stepper motors simultaneously using the **GRBL** library, making it ideal for motion control projects. The module is built with an MEGA328P and comes preloaded with **GRBL** firmware, communicating with the M5Core via I2C(0x70). It integrates 3 stepper motor driver boards based on the DRV8825 chip, a simple yet powerful circuit board that can control a bipolar stepper motor and allows microstepping up to 1/32 steps.

Features

- Power Input: 9 ~ 24V
- Controls 3 stepper motors **(X, Y, Z)**

Includes

- 1 x Module Stepmotor

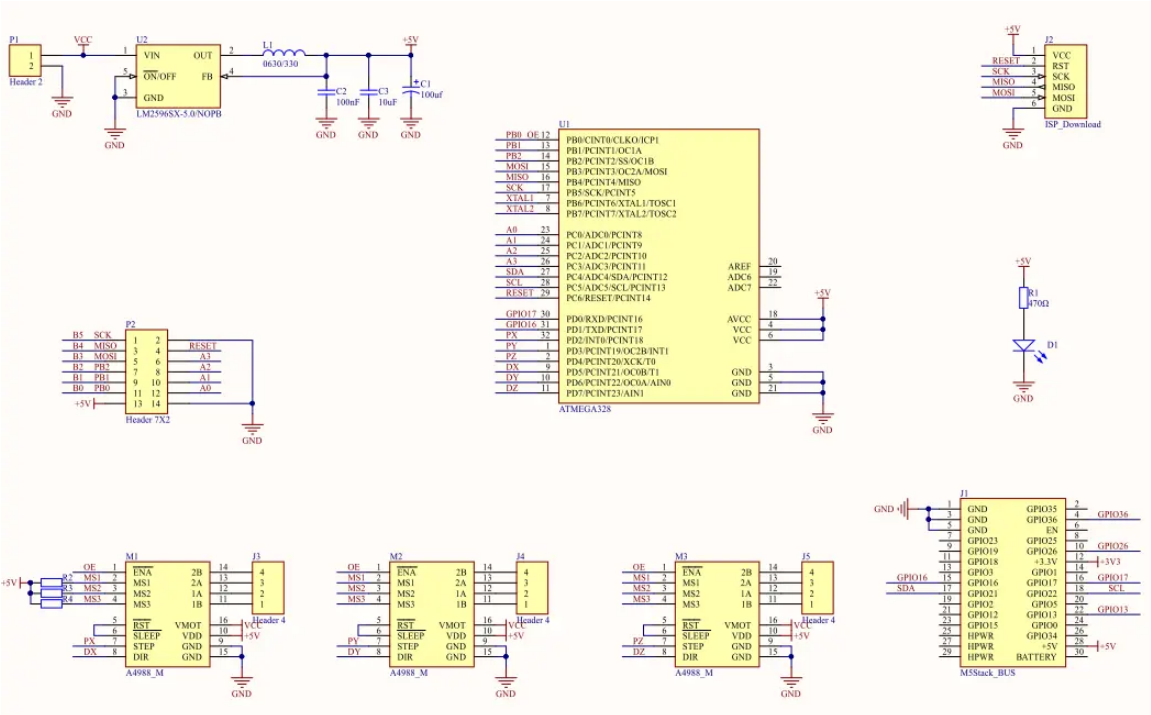
Applications

- DIY 3D Printer
- Building Robotic Arms

Specifications

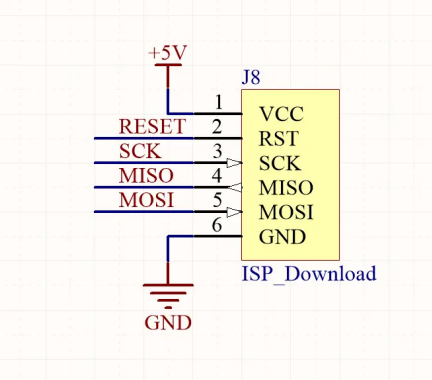
Specification	Parameter
DC Connector Type	XT30
Communication Interface	I2C Communication @0x70
Motor Connector Spec	HY2.0-4P
Net Weight	24g
Gross Weight	34g
Product Size	54.2 x 54.2 x 12.8mm
Package Size	60 x 57 x 17mm

Schematics



PinMap

Mega328 ISP Download Interface Pin Definition



M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	RS485_TX
	9	10	
	11	12	3V3
	13	14	
STEP_X	15	16	DIR_X
SDA	17	18	SCL
	19	20	
STEP_Y	21	22	DIR_Y
STEP_Z	23	24	DIR_Z
HPWR	25	26	RS485_RX
HPWR	27	28	5V
HPWR	29	30	

Datasheets

- DRV8825 Datasheet

Softwares

Arduino

- Module Stepmotor Example with M5Core

UiFlow1

- Module Stepmotor UiFlow1 Docs

Built-in Firmware

- Module Stepmotor Internal Firmware

Easyloader

Easyloader	Download Link	Notes
Module Stepmotor Test Easyloader	download	/